João Lucas **Mayrinck D'Oliveira**

Technology & Software Specialist

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Profile

Researcher and software developer connecting computing, education, and social impact. BSc in Information Systems from UFV (2019–2025) and MSc student in Computer Science at DCC/UFMG (2026–present). Research focus on computer networks, information security, intelligent systems, educational games, and software engineering.

Professional Experience

Federal University of Minas Gerais (DCC / UFMG)

Graduate Researcher / M.Sc. Student — *2026 - Present*

- Research and advanced coursework in computer networks, information security, intelligent systems, and machine learning.

Federal University of Minas Gerais (DCC / UFMG)

Student Electoral Board Member — *2026*

- Participation in the student representation election process for the Graduate Student Academic Center (CAPosComp/UFMG).

Federal University of Viçosa (UFV · Game Dev)

Introduction to Game Development Teaching Assistant — *2025*

- Hands-on guidance and student mentoring, supporting the development and review of course materials in Digital Game Development.

Federal University of Viçosa

Research Fellow — *2023 - 2024*

- Development of dynamic interactive figures in GeoGebra for mathematics education and academic outreach.

Health Lake Institute

Frontend Developer (Volunteer) — *2021 - 2023*

- Volunteer work (Dec 2021 – Mar 2023) in Science and Technology: migrating website to WordPress, creating wireframes using Figma, website administration and maintenance.
- Volunteer work in Science and Technology, supporting academic events and educational workshops.

Federal University of Viçosa

Programming Teaching Assistant — *2021 - 2022*

- Student mentoring and instructional support for learning programming fundamentals and data structures.

Education

Federal University of Minas Gerais (UFMG · DCC)

M.Sc. in Computer Science — *2026 - Present*

Thesis: Advanced studies in wireless networks, security, algorithms, and machine learning

Federal University of Viçosa

B.Sc. in Information Systems — *2019 - 2025*

Thesis: Education in software and systems development, UX design, data analysis, and educational technology

Featured & Open Source Projects

latex-resume

- Automatic generator for Academic and Professional resumes in LaTeX + PDF (supporting both en-US and pt-BR), powered by Multiple AI Providers (NVIDIA NIM, Google Gemini, Anthropic Claude, OpenAI, DeepSeek)...
- **Technologies & Tools:** Python, JavaScript, LaTeX, HTML5, CSS3, Bash

jluckmay.github.io

- Personal and academic portfolio built as a static website.
- **Technologies & Tools:** HTML5, Python, JavaScript, LaTeX, CSS3, Bash

minecraft-map-convertion

- While existing terrain tools (such as Chunker or Amulet) convert blocks, biomes, containers, and dimensions, they do not convert entities, player inventories, custom villager NBT trades, or datapack logic.
- **Technologies & Tools:** Python, Java, Bash, Git

C-Codes

- By João Lucas Mayrinck, Information Systems Bachelor by the Federal University of Viçosa (UFV).

- **Technologies & Tools:** C, JavaScript, GitHub Pages

Quadro-Branco

- Collaborative Whiteboard is a multi-user web application for creating and editing visual content simultaneously.
- **Technologies & Tools:** JavaScript, Python, HTML5, CSS3, SQL, FastAPI

MineMan

- MineMan is a maze exploration game inspired by the Minecraft universe.
- **Technologies & Tools:** HTML5, Python, C++, Bash

Scientific Publications

1. R Coêlho, F Bastos, JL Mayrinck, PM de Sousa. **Math-0-Ween: Um Jogo Digital Educacional para Explorar Operações Matemáticas de Forma Divertida e Assustadora-Resumo Estendido**. *Operações Matemáticas de Forma Divertida e Assustadora Revista Brasileira de Informática na Educação*, 2026. DOI: 10.5753/rbie.2026.6599
2. Abordagem Espacial e Frequencial Anais do XVI Workshop de Sistemas de Informação. **PDI Interativo: Uma Ferramenta para Estudo e Experimentação com Abordagem Espacial e Frequencial**. *com Abordagem Espacial e Frequencial Anais do XVI Workshop de Sistemas de Informação*, 2025. DOI: 10.5753/wsis.2025.15098
3. PL Mariano, F Bastos, JL Mayrinck, PM Sousa. **Jogo Digital Educativo como Ferramenta de Apoio no Ensino de Frações**. *XVI Workshop de Sistemas de Informação – UFV Rio Paranaíba Campus*, 2025. DOI: 10.5753/wsis.2025.15113
4. JL Mayrinck. **Impacto da Solução Inicial na Otimização por Simulated Annealing: Uma Análise Comparativa entre GRASP e HPG**. *Annealing: Uma Análise Comparativa entre GRASP e HPG Anais do XVI Workshop de Sistemas de Informação*, 2025. DOI: 10.5753/wsis.2025.15099
5. JVN Silva, F Bastos, JL Mayrinck, PM de Sousa. **Gamificação no Ensino de Física: Projeto e Teste de Usabilidade de um Jogo de Física para Ensino Médio**. *Usabilidade de um Jogo de Física para Ensino Médio XVI Workshop de Sistemas de Informação*, 2025. DOI: 10.5753/wsis.2025.15027
6. ME Matos, BT de Miranda, F Bastos, JL Mayrinck, A Oda, PM de Sousa. **Explorando a Grécia Antiga: Um Jogo Digital Educativo**. *XVI Workshop de Sistemas de Informação*, 2025. DOI: 10.5753/wsis.2025.15623
7. **Atividades de aprendizagem em matemática baseadas no uso interativo de figuras dinâmicas em ambientes virtuais**. *Autoria, tecnologia e direitos humanos na educação: diálogos com Pedro Demo*, 2025.
8. JL Mayrinck, F Bastos, A Oda, AC Inocêncio, MA Névoa, MW Ribeiro, ... **Palavras ao Ar: um jogo digital de descoberta de novas palavras e seus significados para auxiliar na alfabetização**. *Simpósio Brasileiro de Informática na Educação (SBIE), 591-601 , 2024*, 2024. DOI: 10.5753/sbie.2024.241887
9. **Palavras ao Ar: um jogo digital de descoberta de novas palavras e seus significados para auxiliar na alfabetização**. *XIV Workshop de Sistemas de Informação – UFV Rio Paranaíba Campus*, 2023.

Awards, Honors & Certifications

- Honorable Mention – Poster Track – WSIS 2025 (Explorando a Grécia Antiga)
- Honorable Mention – Games Track – SBIE 2024 (Palavras ao Ar)
- Second Place – Teaching & Education – SIA 2023 (Palavras ao Ar)

Languages

Portuguese ●●●●●

English ●●●○○○

Spanish ●●○○○○

Libras ●○○○○○